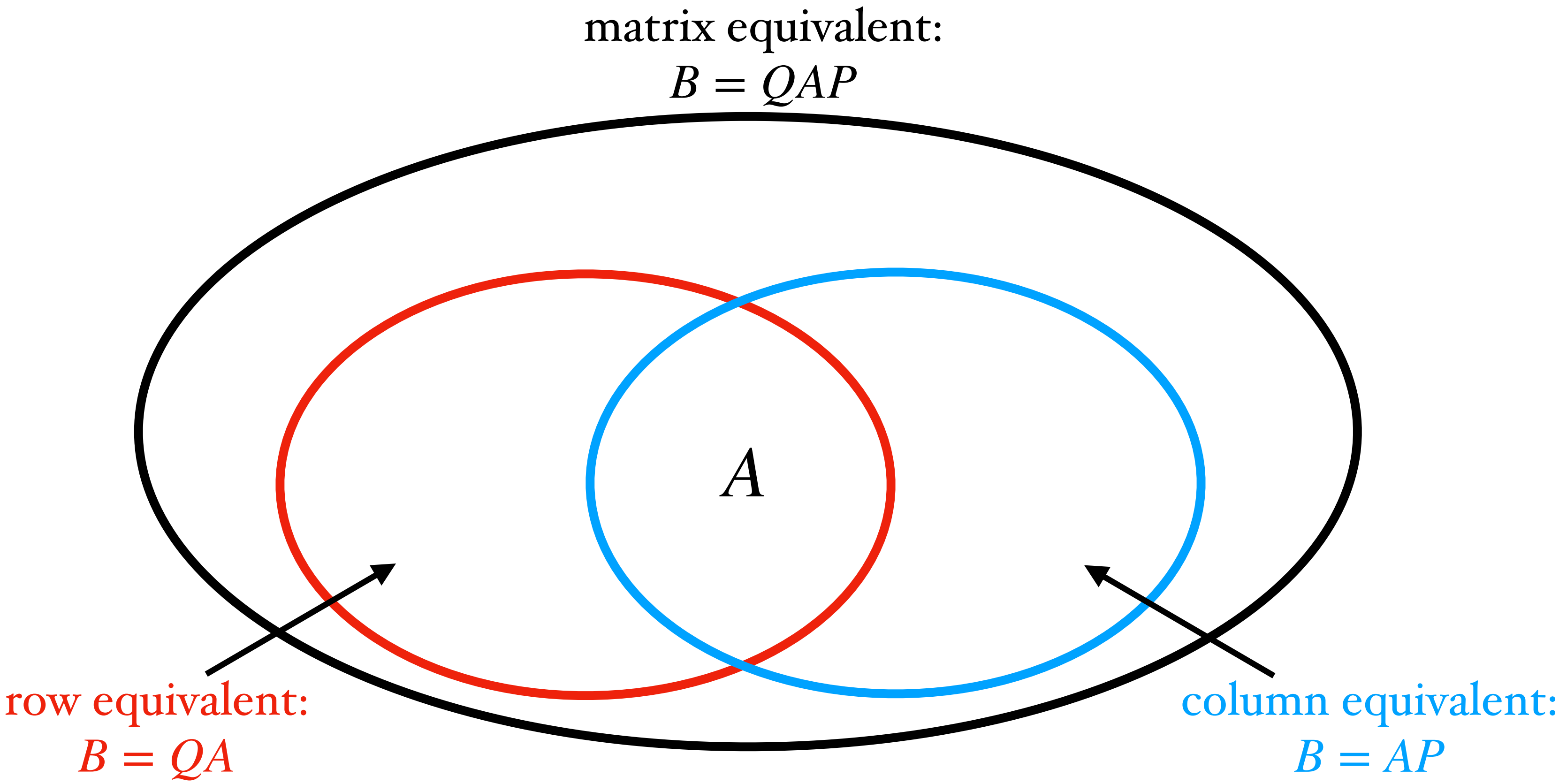
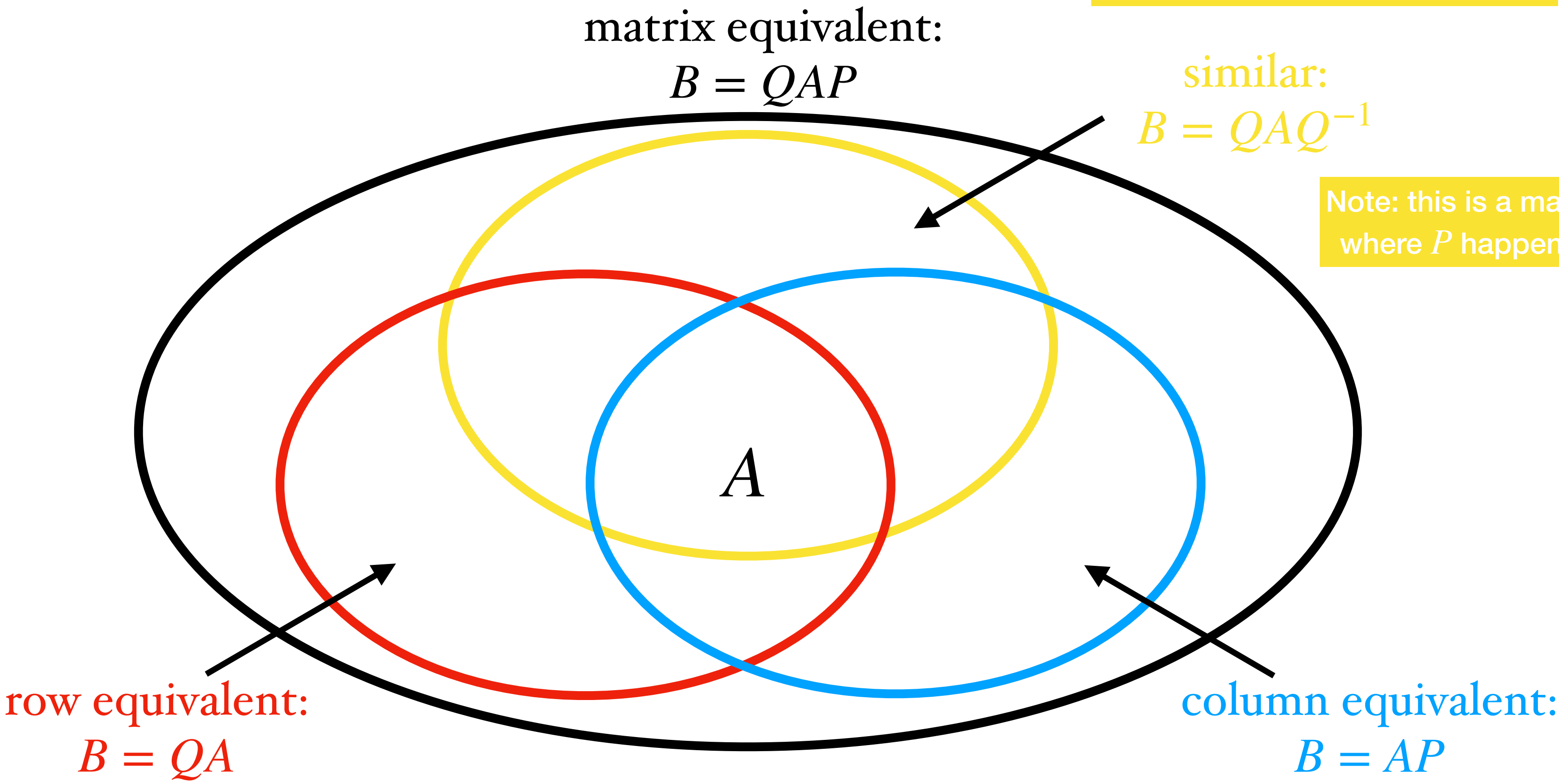
For an arbitrary $A \in M^{\mathbb{F}}_{m\times n}$, we have the following containments among equivalence relations.
In each case, B is equivalent to A if it satisfies a certain equation relating A and B.
(Here Q is an arbitrary invertible $m \times m$ matrix, and P is an arbitrary invertible $n \times n$ matrix.)



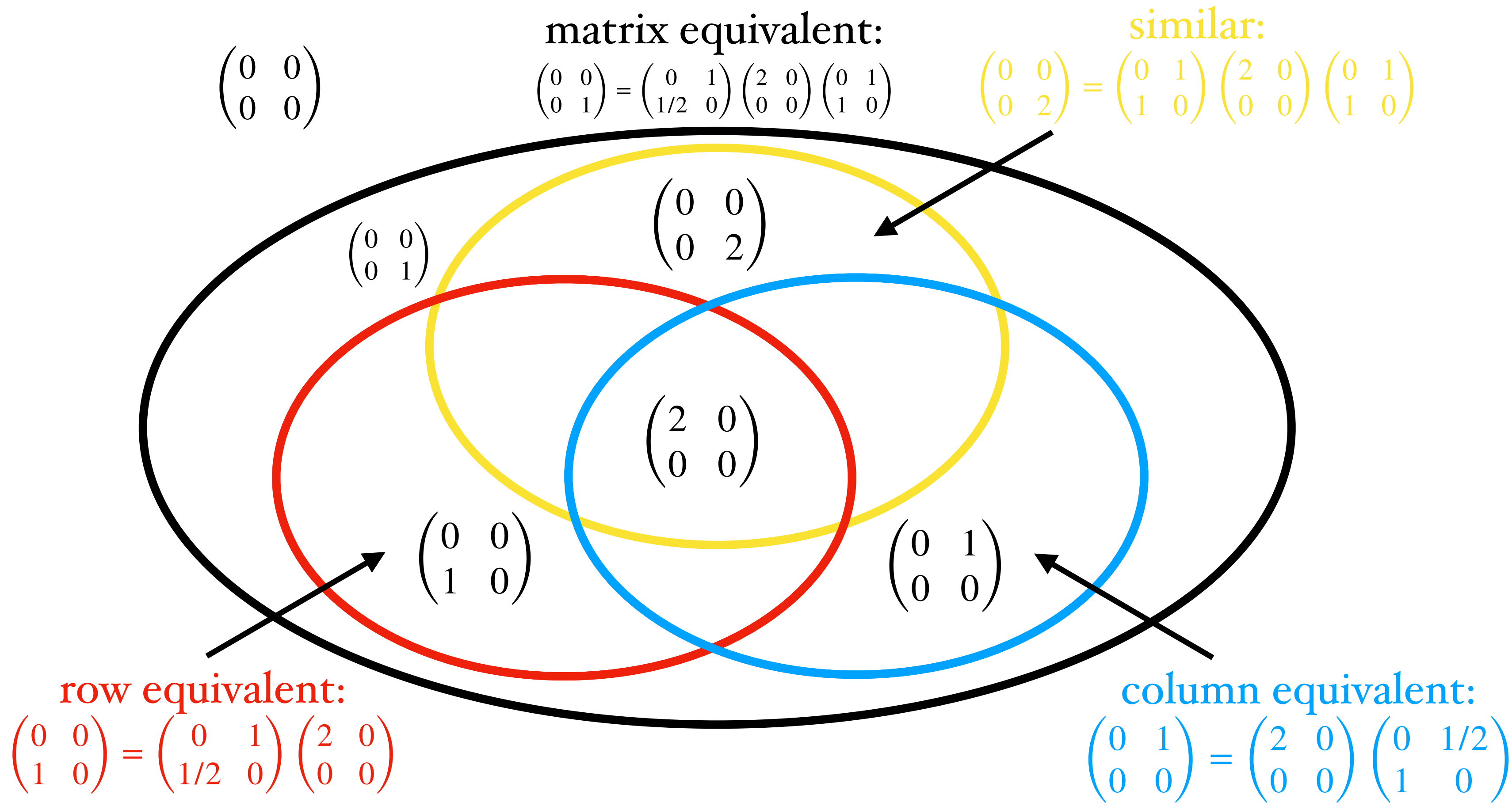
For an arbitrary $A \in M^{\mathbb{F}}_{n\times n}$, we have the following containments among equivalence relations.
In each case, B is equivalent to A if it satisfies a certain equation relating A and B.
(Here Q is an arbitrary invertible $n \times n$ matrix.)

Note: similarity only makes sense for square matrices.



Note: this is a matrix equivalence, where P happens to equal Q^{-1}.

For the matrix $A \in M^{\mathbb{R}}_{2\times 2}$ that is $A = \begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix}$, we have the following examples:



For the matrix $A \in M^{\mathbb{R}}_{2\times 2}$ that is $A = \begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix}$, we have the following examples: